You should have done it based on some properties. Brick and door which are hard come under solids, things that flow come under liquids and others which are very light and can flow more freely come under gases.

Activity 3

Malar was asked to group some items based on their physical states. The table she made is given below. Do you agree with her? Correct the table if you do not agree and submit it to your teacher. (Work in a group of two.)

Chalk piece	Wind	Steam
Water	Rain	Lemon
Air in a balloon	Stone	Lemon juice
River	Air	Smoke
Brick	Table	Door

3.3 Diffusion

Let us place a book on a table. Let it not be disturbed. Observe for five minutes. Now take a glass of water and add a drop of ink carefully at the centre. Do not shake or stir.

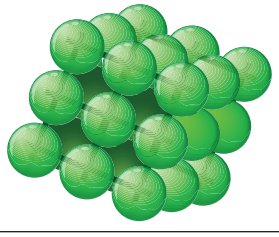
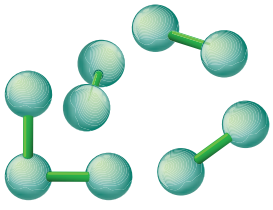
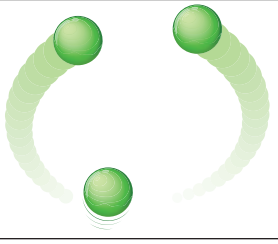
Now light an incense stick and keep it in one corner of the room.



Let us answer the following questions.

- Did the book move?
- Did the ink particles move and spread itself in the water? How long did it take for complete mixing?
- Did you get the smell of the incense stick from where you are standing?
- How fast did you get the smell? How did the smell reach you?

We may conclude that the particles of gases and liquids can move easily and quickly. This tendency of particles to spread out in order to occupy the available space is called diffusion. Solids are tightly packed and they do not diffuse like liquids or gases. Hence ink and smoke spread easily while book stays on the table.

Particles in a Solid	Particles in a Liquid	Particles in a Gas
		
In solid, the particles are tightly packed with very little space between them. Eg. Stone	Particles in liquids are arranged in a random or irregular way and the space between the particles is greater than that is in solids. Eg. Water	The particles in the gases are arranged far apart. They move freely. Eg. Air